Attitude Dynamics
Quaternions

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1 Representing Rotations

The modeling of attitude dynamics is an absolutely indispensable part of space mission design. However, the rotation of objects in three dimensions, around multiple principal axes simultaneously and subject to unbalanced torques, is notorious for causing even the most capable engineers nearly incurable headaches. Eventually, engineers tired of taking aspirin for their attitude headaches, so they starting coming up with standard conventions for representing attitude. However, each of these conventions brought along their own causes for frustration.

Note: in this document, and throughout *Star Chart*, atittude representations will correspond to rotations *from* the body frame of a satellite *to* a relevant inertial frame (such as ECI).

- **Euler Angles**. These are... *awful*. There are no fewer than *literally* 24 competing conventions for Euler angles. Also, each unique attitude state has a countably infinite number of Euler angles corresponding to it (because you can add 2π radians or 360 degrees to wrap around).

- **Rotation Matrices**. These are a bit better than Euler angles. One rotation matrix exists for each attitude state, unlike Euler angles. However, they come at a cost... they're extremely overparameterized. Attitude is inherently a "three-dimensional" thing, but rotation matrices require nine values (square 3×3 matrix). This requires lots of memory when doing complex calculations. They also encounter significant numerical issues during estimation tasks, such as least-squares estimation.

- **Axis-Angle Vectors**. Axis-angle vectors, like Euler angles are minimal in their parameterization of attitude, as they only require three values. However, they too experience suffer from the "wrap-around" problem of Euler angles. Because the length of an axis-angle vector corresponds to the rotation angle (typically in radians), adding 2π to the length of an axis-angle vector constructs a distinct vector representing the exact same attitude state.

For aforementioned reasons, none of these parameterizations have reached widespread adoption for attitude dynamics, and those who have chosen to adopt one of these has struggled immeasurably with the unique issues corresponding to each. The **quaternion** representation of attitude, unlike these, is unproblematic numerically, has only two competing conventions, and minimizes overparameterization.

2 Quaternions

The vector space of **quaternions** is a natural extension of the complex numbers, and quaternions can be used to naturally represent rotations. A quaternion consists of a *scalar part* s and a *vector part* $\mathbf{v}$.

$$q = \begin{bmatrix} s \\ \mathbf{v} \end{bmatrix} \tag{1}$$

Here, we will always write the scalar part *before* the vector part. Some people write the scalar part after the vector part; this is the only other competing convention.